

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The network protocol analyzer logs reveal that port 53, typically used for DNS services, is unreachable when attempting to access the website www.yummyrecipesforme.com. This issue was detected via ICMP error messages, specifically indicating "udp port 53 unreachable.". This may indicate a problem with the DNS server or the firewall configuration. Additionally, the incident could potentially indicate a malicious attack targeting the DNS server, such as a Denial-of-Service (DoS) attack or other network-based exploits.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred around 1 PM, as reported by several customers who were unable to access the client company website, www.yummyrecipesforme.com. Users received the error message "destination port unreachable" when attempting to load the website. The network security team responded and began running tests with the network protocol analyzer tool tcpdump. The resulting logs revealed that port 53, which is used for DNS service, is not reachable. The DNS server could be down, misconfigured, or its DNS service might not be running or a firewall or other security mechanism might be blocking UDP traffic to port 53.